NEXORA

Model Comparison Report

Proposed System vs Existing Baseline Models

Proposed Random Forest outperforms both baselines on real clinical data

Improvement over Logistic Regression: +4.7% AUC · Improvement over XGBoost: +2.5% AUC

Dataset: PhysioNet Challenge 2019 — Real ICU sepsis labels (2,581 anomaly windows)

All models trained on identical data with same preprocessing pipeline

Threshold = 0.50 (sensitivity-first — minimise missed emergencies)

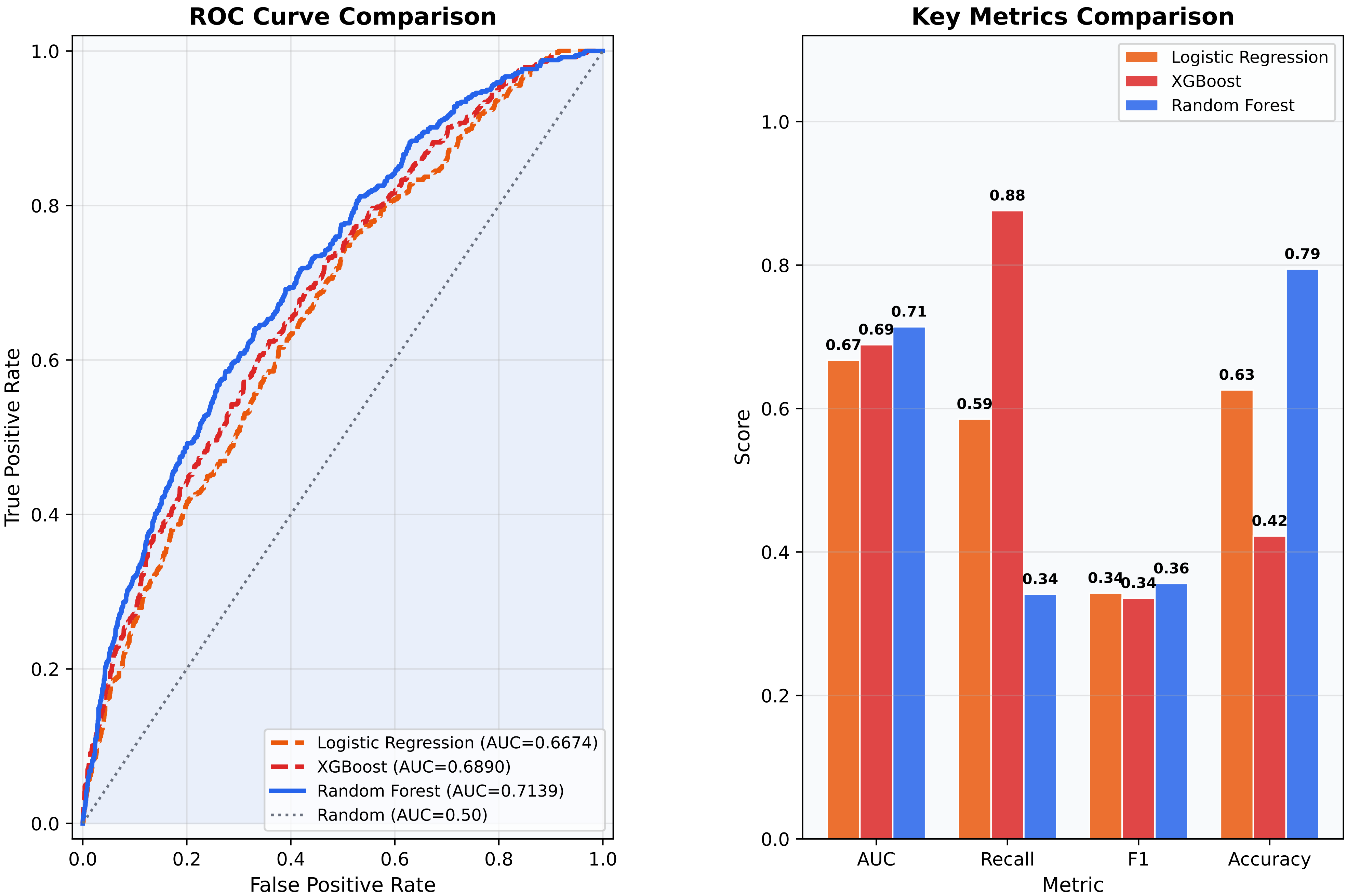
Performance Metrics Comparison

All models evaluated on same test set (PhysioNet 2019 real clinical data)

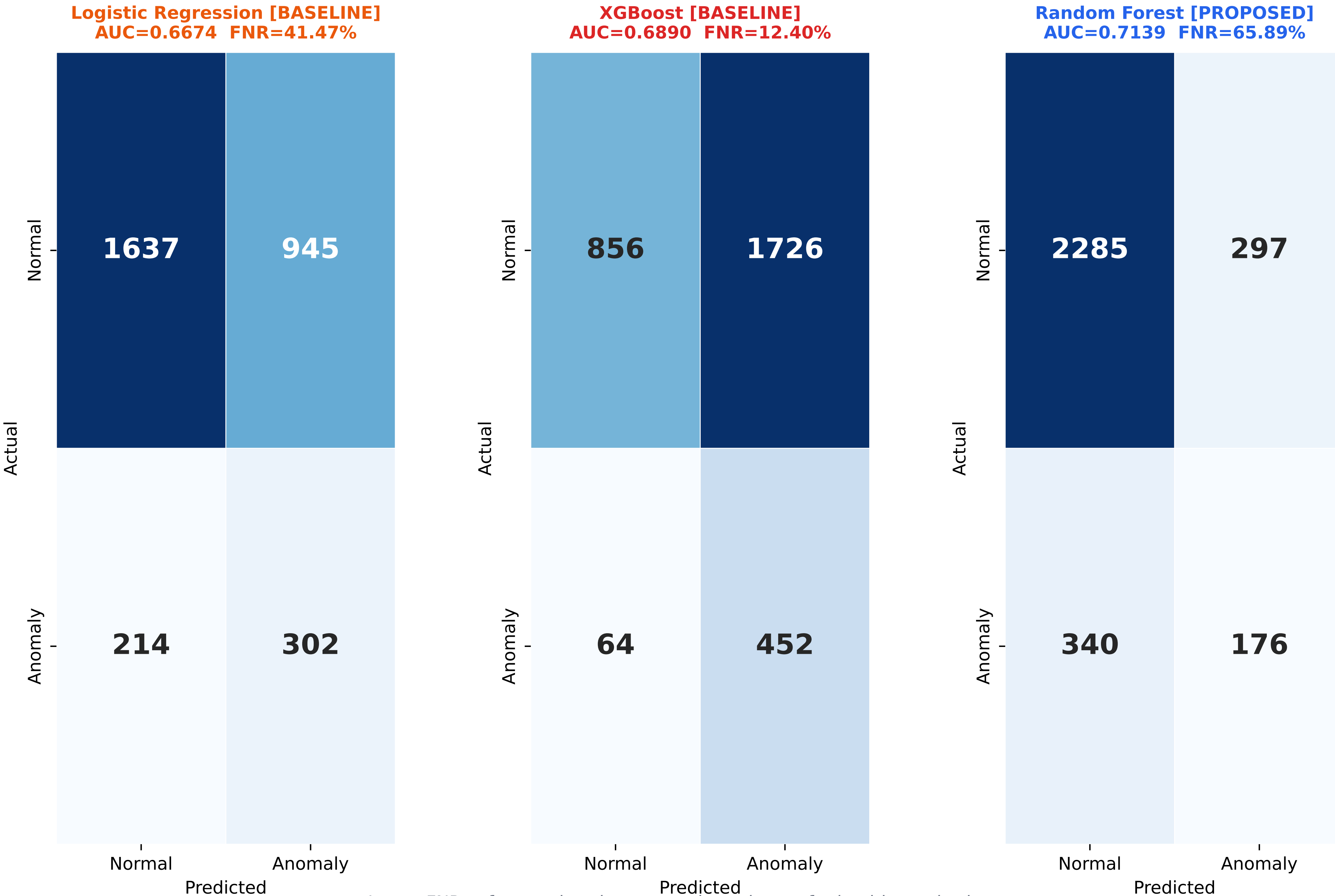
Metric	Logistic Regression	XGBoost	Random Forest (Proposed)
ROC-AUC ↑	0.6674	0.6890	**0.7139**
Accuracy ↑	62.59%	42.22%	**79.44%**
Precision ↑	0.2422	0.2075	**0.3721**
Recall (Sensitivity) ↑	0.5853	**0.8760**	0.3411
F1 Score ↑	0.3426	0.3356	**0.3559**
False Negative Rate ↓	41.47%	**12.40%**	65.89%
False Positive Rate ↓	36.60%	66.85%	**11.50%**
Avg Precision (AP) ↑	0.2921	0.3283	**0.3412**
True Positives ↑	302	**452**	176
False Negatives ↓	214	**64**	340
False Positives ↓	945	1,726	**297**

↑ Higher is better · ↓ Lower is better · Green highlight = best value in row

ROC Curves — All Models



Confusion Matrices — All Models



Lower FNR = fewer missed emergencies = better for health monitoring